

2212 Java Software Dev I

Project 1 – Calorie Burn

Now that you've installed your chosen Integrated Development Environment (IDE), it's time to start coding. Create a new Java project and write code to solve the problem described below. The skills needed for this task are covered in Units 1 and 2 of the course.

Calorie Burn Project Overview

Purpose

The purpose of this project is to practice using an Integrated Development Environment (IDE) to create a Java program that utilizes input statements, variable declarations, assignment statements, calculations, and output statements.

Goal

The goal is to write a program that gathers information about the user's exercise session and displays the estimated number of calories burned.

Calorie Burn Calculation

According to a group of researchers, the amount of energy burned during exercise, measured in calories, can be estimated using a specific formula.

$$\text{Calories} = \frac{(\text{Age} \times 0.2757 + \text{Weight} \times 0.03295 + \text{HeartRate} \times 1.0781 - 75.4991) \times \text{Time}}{8.368}$$

Program Requirements

The program should:

- Request the user's name and the four input values necessary for the calculation.
- Use the provided formula to calculate the amount of energy used in an exercise session.
- Store the calculation result in a variable.
- Output a message that includes the user's name and the estimated number of calories burned.

Constants to Use

The following constants should be created and used in the calculation:

- Age factor: 0.2757
- Weight factor: 0.03295
- Heart rate factor: 1.0781
- Base offset: 75.4991
- Normalization factor: 8.368

Expectations

The program is expected to compile successfully, include identification comments at the top, and produce output that exemplifies good grammar and spelling. Failure to meet these standards may result in point deductions.